Constraint programming: propagation and search



Optimization · Level 4 · Lesson 3

In this lesson you will

- Describe a problem with domains and constraints, and propagate by hand.
- Explain fail-first search, global constraints and symmetry breaking.
- Model a puzzle in CP-SAT, and read what INFEASIBLE proves.

Key ideas

- Propagation: safe deduction to a fixpoint
- Search: fail-first, undo dead ends
- Global constraints prune more than their pieces
- INFEASIBLE is a proof; UNKNOWN is not

Activity: Be the solver

1. Propagate (8 min): Write candidates and cross out.
2. Fail-first (7 min): Guess where the fewest remain.
3. Count (5 min): Compare your guesses with a partner’s.
4. Model (5 min): Write the same rules as CP-SAT lines.

Any sudoku from a newspaper works. A pencil and an eraser make undoing guesses easy.

Let’s discuss

1. Why is fail-first a good rule, even though it guesses where you are least sure?
2. Which real problems have symmetry worth breaking?
3. How would you explain an INFEASIBLE roster to the people who wrote its rules?

Check yourself

- Explain in two sentences why all-different is stronger than its 36 pairwise rules.